ON QCQS BASES EVERY FPPF COVERING IS DOMINATED BY A SINGLE FINITE FINITELY PRESENTED SURJECTION AFTER OPEN REFINEMENT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 13757-13765 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 13767-13785, the complete author proof (17 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The finiteness upgrade in the chapter's domination trio; sequenced here after its users so the shelf is complete (L16098 cites it, L16107's etale-local variant stands beside it).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemma dominate-fppf-integral (the integral domination); Limits, lemma integral-limit-finite-and-finite-presentation, lemma descend-opens, and proposition characterize-locally-finite-presentation (the finite-presentation limit machinery).

Source reference. Original lemma 0CNZ: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13726–13731. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01Z4: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13785–13785. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 09YZ: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13785–13785. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01ZC: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13785–13785. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a quasi-compact and quasi-separated scheme. Let $\{S_i \to S\}_{i \in I}$ be an fppf covering. Then there exists a surjective finite morphism $S' \to S$ of finite presentation and an open covering $S' = \bigcup U'_\alpha$ such that for each α the morphism $U'_\alpha \to S$ factors through $S_i \to S$ for some i.*

Proof. Let $Y \to X$ be the integral surjective morphism found in Lemma 0CNZ. Choose a finite affine open covering $Y = \bigcup V_j$ such that $V_j \to X$ factors through $S_{i(j)}$. We can write $Y = \lim Y_\lambda$ with $Y_\lambda \to X$ finite and of finite presentation, see Limits, Lemma 09YZ. For large enough λ we can find affine opens $V_{\lambda,j} \subset Y_\lambda$ whose inverse image in Y recovers V_j, see Limits, Lemma 01Z4. For even larger λ the morphisms $V_j \to S_{i(j)}$ over X come from morphisms $V_{\lambda,j} \to S_{i(j)}$ over X, see Limits, Proposition 01ZC. Setting $S' = Y_\lambda$ for this λ finishes the proof. □